Cinélux

Cinema Ticket Booking System

Finalized Requirements Specification (SRS)

Expert SRS Reviewer

March 26, 2026

Table 1: Functional Requirements (21 items)

ID	Type	Category	Priority	Details
FR1	Functional	Auth	Must Have	User shall register an account using a unique email address and password.
FR2	Functional	Auth	Must Have	User shall log into the system using valid email credentials.
FR3	Functional	Auth	Must Have	User shall be able to initiate a password reset via a verified link sent to their email.
FR4	Functional	Content	Must Have	System shall display movie details: title, genre, duration, description, and poster.
FR5	Functional	Booking	Must Have	User shall select a specific movie from the available catalog for reservation.
FR6	Functional	Booking	Must Have	User shall select an available showtime date and hour for a chosen movie.
FR7	Functional	Booking	Must Have	User shall select specific available seats from a visual cinema hall seating chart.
FR8	Functional	Booking	Must Have	User shall confirm the booking details to finalize the reservation process.
FR9	Functional	Profile	Should Have	User shall select preferred movie genres during the initial account onboarding process.
FR10	Functional	Booking	Must Have	System shall place a 10-minute temporary hold on selected seats during checkout.
FR11	Functional	Social	Should Have	User shall submit a movie rating on a scale of 1 to 5 stars.
FR12	Functional	Pricing	Must Have	System shall display showtime pricing based on selected seat type (e.g., VIP, Standard).
FR13	Functional	History	Should Have	User shall view a list of all their past and upcoming movie ticket bookings.
FR14	Functional	Notification	Must Have	System shall send a booking confirmation receipt and digital ticket to the user's email.
FR15	Functional	Search	Should Have	User shall search for movies by title or filter the catalog by primary genre.
FR16	Functional	Personalization	Should Have	System shall recommend movies based on the user's selected genre preferences.
FR17	Functional	Personalization	Should Have	System shall recommend movies based on the user's historical booking data.
FR18	Functional	AI Features	Should Have	System shall provide AI-generated insights on trending movies within the current region.
FR19	Functional	AI Features	Should Have	System shall suggest optimal showtimes based on previous user activity patterns.
FR20	Functional	AI Features	Should Have	System shall offer personalized movie discounts predicted to interest the specific user.
FR21	Functional	Management	Must Have	Admin shall add, update, or remove movie listings and associated showtime schedules.

Table 2: Non-Functional Requirements (8 items)

ID	Type	Category	Priority	Details
NFR1	Non-Func	Performance	Must Have	System shall load the primary movie catalog page within 2.0 seconds.
NFR2	Non-Func	Performance	Must Have	System shall support 10+ concurrent users during the demo without degradation.
NFR3	Non-Func	Reliability	Must Have	System shall remain stable and operational for the entire duration of the presentation.
NFR4	Non-Func	Usability	Must Have	Application shall be responsive and usable on both mobile and desktop web browsers.
NFR5	Non-Func	Security	Must Have	System shall encrypt all user passwords using the BCrypt hashing algorithm.
NFR6	Non-Func	Security	Must Have	System shall automatically terminate user sessions after 30 minutes of inactivity.
NFR7	Non-Func	Maintainability	Should Have	Code shall use modular components to allow genre list updates within 24 hours.
NFR8	Non-Func	Compatibility	Must Have	System shall maintain full functionality on the latest versions of Chrome and Safari.

REQUIREMENTS TRACEABILITY MATRIX (RTM)

Functional Requirements Only | FR1 – FR21

ID	Requirement Title	Use Case	Analysis	Sys Design	Det Design	Coding	UI	Test Int
FR1	User Registration – Creating a new user account with validation and secure data handling	UC01 – Register Account						
FR2	User Login – Secure authentication using verified user credentials	UC02 – Login						
FR3	Browse Movies – Viewing all available movies with posters and basic details	UC03 – Browse Movies						
FR4	View Movie Details – Accessing full movie information including cast, trailer, and showtimes	UC04 – View Movie Details						
FR5	Make a Booking – Selecting movie, showtime, and seat to complete reservation	UC05 – Make Booking						
FR6	Generate QR Ticket – Creating a unique, scannable QR code for each booking	UC06 – Generate QR Ticket						
FR7	Booking Notification – Sending confirmation message with booking details and QR code	UC07 – Send Booking Notification						
FR8	Cancel Booking – Allowing users to cancel bookings and update seat availability	UC08 – Cancel Booking						
FR9	Cancellation Notification – Sending notification after booking cancellation	UC09 – Send Cancellation Notification						
FR10	View Booking History – Displaying past and upcoming bookings with details	UC10 – View Booking History						
FR11	Add Movie – Admin adds new movies with full metadata and poster	UC11 – Add Movie						
FR12	Edit Movie – Admin updates existing movie information and media	UC12 – Edit Movie						
FR13	Delete Movie – Admin removes movies from the system permanently	UC13 – Delete Movie						
FR14	Manage Showtimes – Admin creates and edits showtimes for each movie	UC14 – Manage Showtimes						
FR15	Manage Halls – Admin manages hall capacity, layout, and seat configuration	UC15 – Manage Halls						
FR16	View Reports – Admin views booking statistics and performance reports	UC16 – View Reports						
FR17	Validate Ticket – Staff scans QR code to verify ticket authenticity	UC17 – Validate Ticket						
FR18	Generate Recommendations – System generates personalized movie suggestions	UC18 – Generate Recommendations						
FR19	Send User Data to AI – System sends user preferences to AI engine	UC19 – Send User Data to AI						
FR20	Receive AI Recommendations – System receives and displays AI suggestions	UC20 – Receive AI Recommendations						
FR21	Send Email/SMS Notifications – System sends all system notifications reliably	UC21 – Send Notifications						

Note: Analysis, System Design, Detailed Design, Coding, UI, and Test Integration columns to be completed as project phases progress.

STAKEHOLDERS

CineLux System | All Roles & Responsibilities

👤 End Users

PRIMARY ACTORS

Registered cinema-goers who browse movies, make bookings, manage reservations, and receive QR tickets and notifications.

Related: FR1–FR10, FR18–FR20

⚙ Admin

SYSTEM MANAGER

Manages the full movie catalog, showtimes, and hall configurations. Views operational reports and booking statistics.

Related: FR11–FR16

🎫 Staff

TICKET VALIDATORS

Cinema entrance staff who scan and validate QR codes to verify ticket authenticity and prevent fraudulent entry.

Related: FR17

🤖 AI Engine

EXTERNAL SERVICE

External AI service that receives user preference data and returns personalized movie recommendations to the backend.

Related: FR18–FR20

📨 Notification Service

EXTERNAL PROVIDER

Third-party Email/SMS provider integrated to reliably deliver booking confirmations and cancellation notifications.

Related: FR7, FR9, FR21

💼 Project Sponsors

MANAGEMENT

Business stakeholders and project sponsors who define system goals, approve requirements, and oversee delivery milestones.

Related: All Requirements

CINÉLUX SYSTEM

Detailed Use Cases

Software Analysis Document
Use Case Specification · Academic Format

Use Case 1 · Make Booking
Use Case 2 · Get Movie Recommendations (AI)

USE CASE 1: MAKE BOOKING

Use Case Name:
Make Booking

Primary Actor:
Registered User

Supporting Actors:
Payment Gateway · Notification Service

Goal:
To allow a registered user to select a movie, choose seats, complete payment, and receive a confirmed booking with a QR code.

Preconditions:
1. The user is registered and logged into the Cinélux system.
2. At least one movie session is available for booking.
3. The Payment Gateway is operational.

Postconditions:
1. A booking record is created and stored in the system.
2. A QR code is generated and associated with the confirmed booking.
3. The user receives a booking confirmation notification.

Main Flow:
1. The user searches for an available movie and selects a desired session.
2. The system displays available seats for the selected session.
3. The user selects one or more seats and proceeds to checkout.
4. The system presents the booking summary, including total cost and selected session details.
5. The user selects a payment method and submits payment details.
6. The system forwards the payment request to the Payment Gateway.
7. The Payment Gateway processes the transaction and returns a success response.
8. The system confirms the booking and saves the reservation record.

(Include: Generate QR (Internal Step)) — The system internally generates a unique QR code linked to the confirmed booking record.
(Include: Send Notification (Internal Step)) — The system internally triggers the Notification Service to deliver a confirmation message and QR code to the user via email or SMS.

A1 – Seat No Longer Available:

A1.1. At step 3, if the selected seat has been taken by another user, the system alerts the user that the seat is unavailable.

A1.2. The system refreshes the seat map and prompts the user to select an alternative seat.

A1.3. The use case resumes at step 3.

A2 – User Modifies Seat Selection:

A2.1. At step 3, the user deselects a chosen seat and selects a different one.

A2.2. The system updates the booking summary accordingly.

A2.3. The use case resumes at step 4.

Exception Flows:

E1 – Payment Failure:

E1.1. At step 7, the Payment Gateway returns a failure or decline response.

E1.2. The system notifies the user that the payment was unsuccessful.

E1.3. The booking is not confirmed and no QR code is generated.

E1.4. The system prompts the user to retry with a different payment method or cancel the booking.

USE CASE 2: GET MOVIE RECOMMENDATIONS (AI)

Use Case Name:
Get Movie Recommendations

Primary Actor:
Registered User

Supporting Actors:
AI Recommendation Engine

Goal:
To provide the registered user with personalised movie recommendations generated by the AI Recommendation Engine, based on the user's preferences and viewing history.

Preconditions:
1. The user is registered and logged into the Cinélux system.
2. The user has an existing viewing history or has configured genre/preference settings.
3. The AI Recommendation Engine is available and accessible.

Postconditions:
1. A personalised list of recommended movies is displayed to the user.
2. The AI interaction and recommendation results are logged for future refinement.

Main Flow:

1. The user navigates to the Recommendations section of the Cinélux platform.

2. The system retrieves the user's viewing history and stored preferences from the database.

3. The system packages the user profile data for submission to the AI Recommendation Engine.

> ***(Include: Prepare AI Request (Internal)) — The system internally structures and formats the user profile data into a valid request payload compatible with the AI Recommendation Engine's interface.***

4. The system submits the prepared request to the AI Recommendation Engine.

5. The AI Recommendation Engine processes the request and returns a ranked list of recommended movies.

> ***(Include: Process AI Response (Internal)) — The system internally parses, validates, and maps the AI engine's response to displayable movie records within the Cinélux platform.***

6. The system presents the personalised movie recommendations to the user, including titles, genres, ratings, and available sessions.

7. The user browses the recommendations and optionally selects a movie to view further details or proceed to booking.

Alternate Flows:

A1 – Insufficient User History:

A1.1. At step 2, if the system determines that the user has insufficient viewing history or no preferences configured, it substitutes general popularity-based data as the input profile.

A1.2. The use case resumes at step 3 using the substituted data.

A1.3. The system displays a notice informing the user that recommendations are based on general trending content.

Exception Flows:

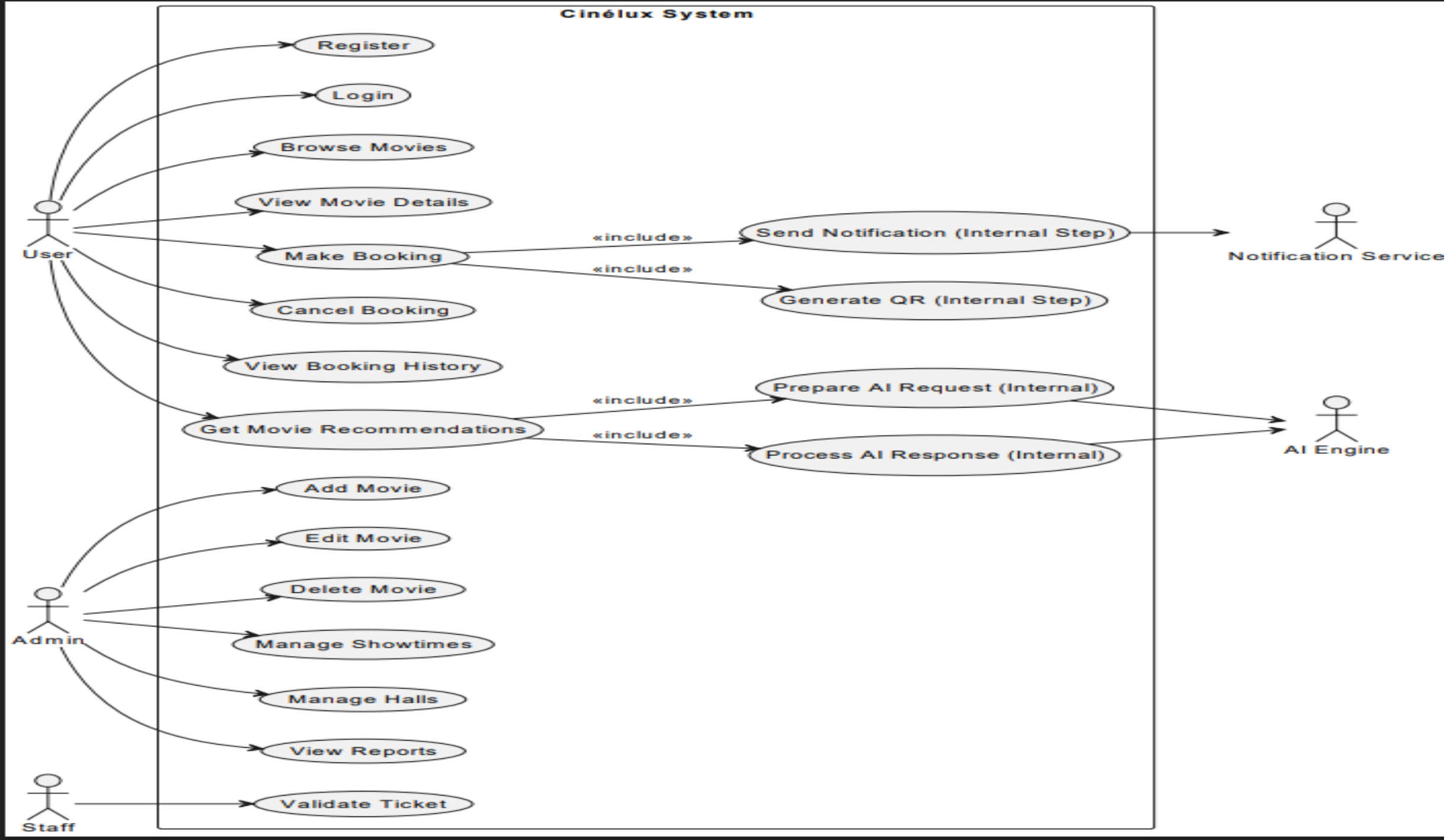
E1 – AI Recommendation Engine Unavailable:

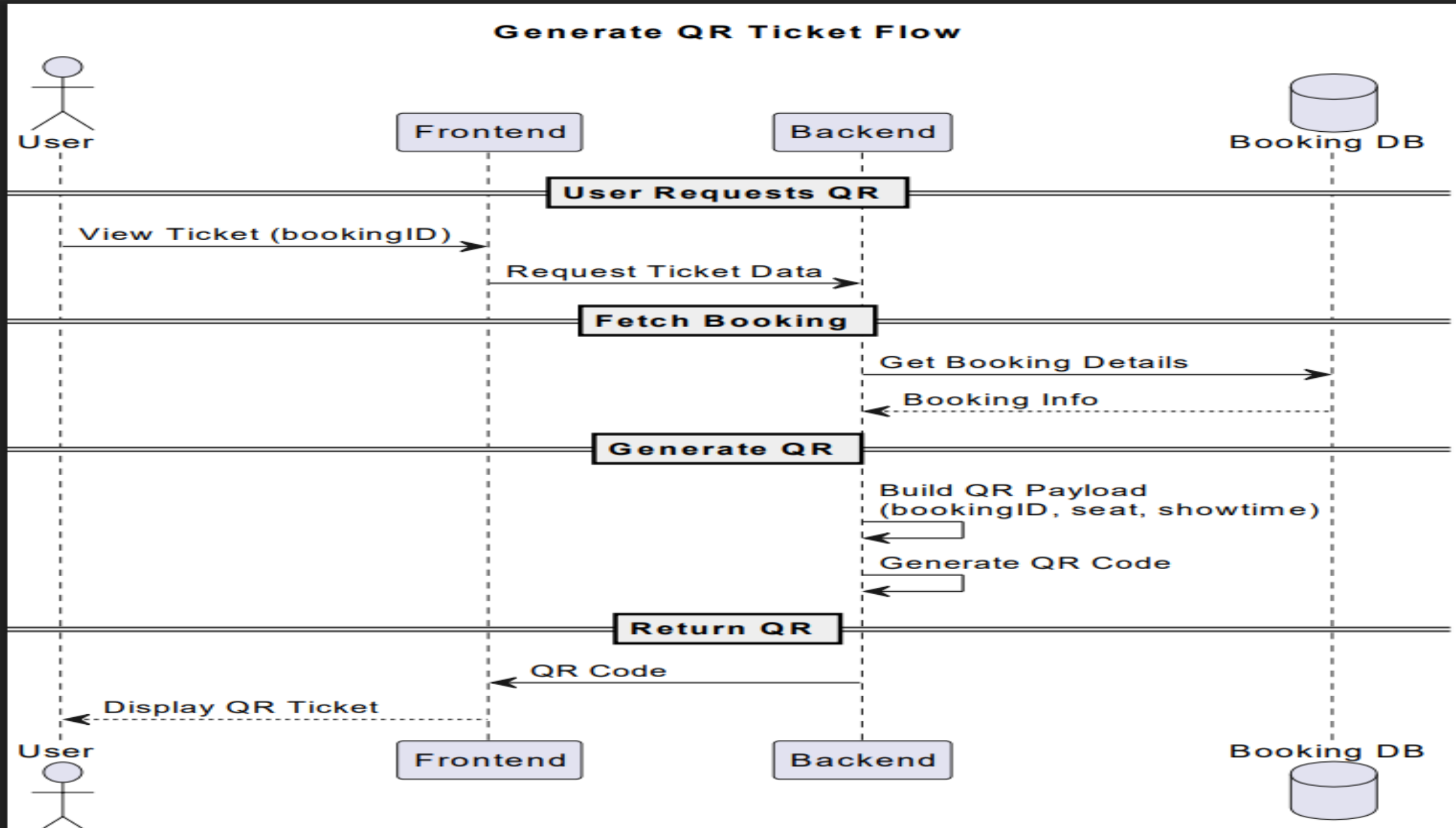
E1.1. At step 4, the system fails to establish a connection with the AI Recommendation Engine or receives a timeout response.

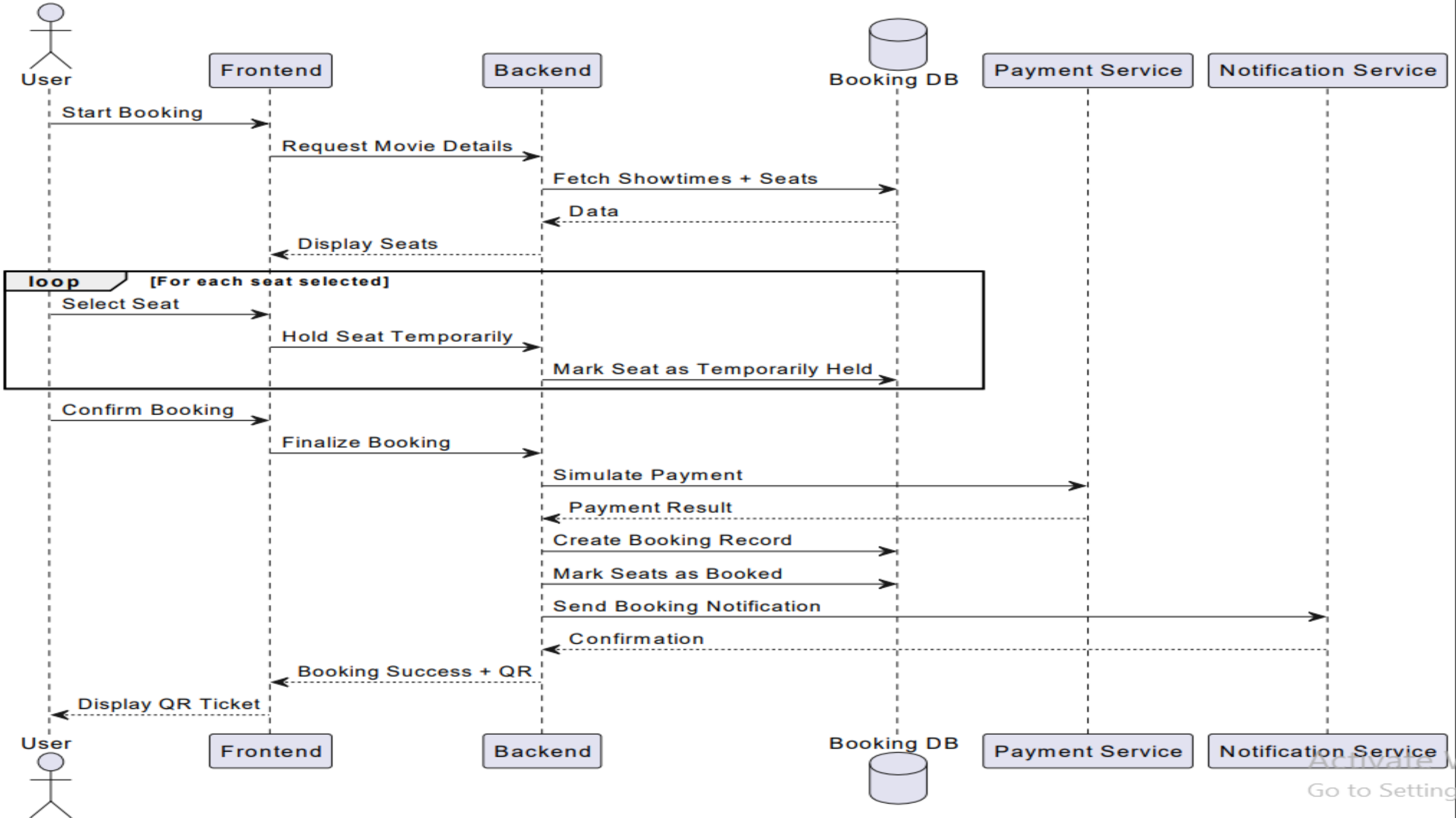
E1.2. The system logs the error and does not proceed with AI-based recommendation processing.

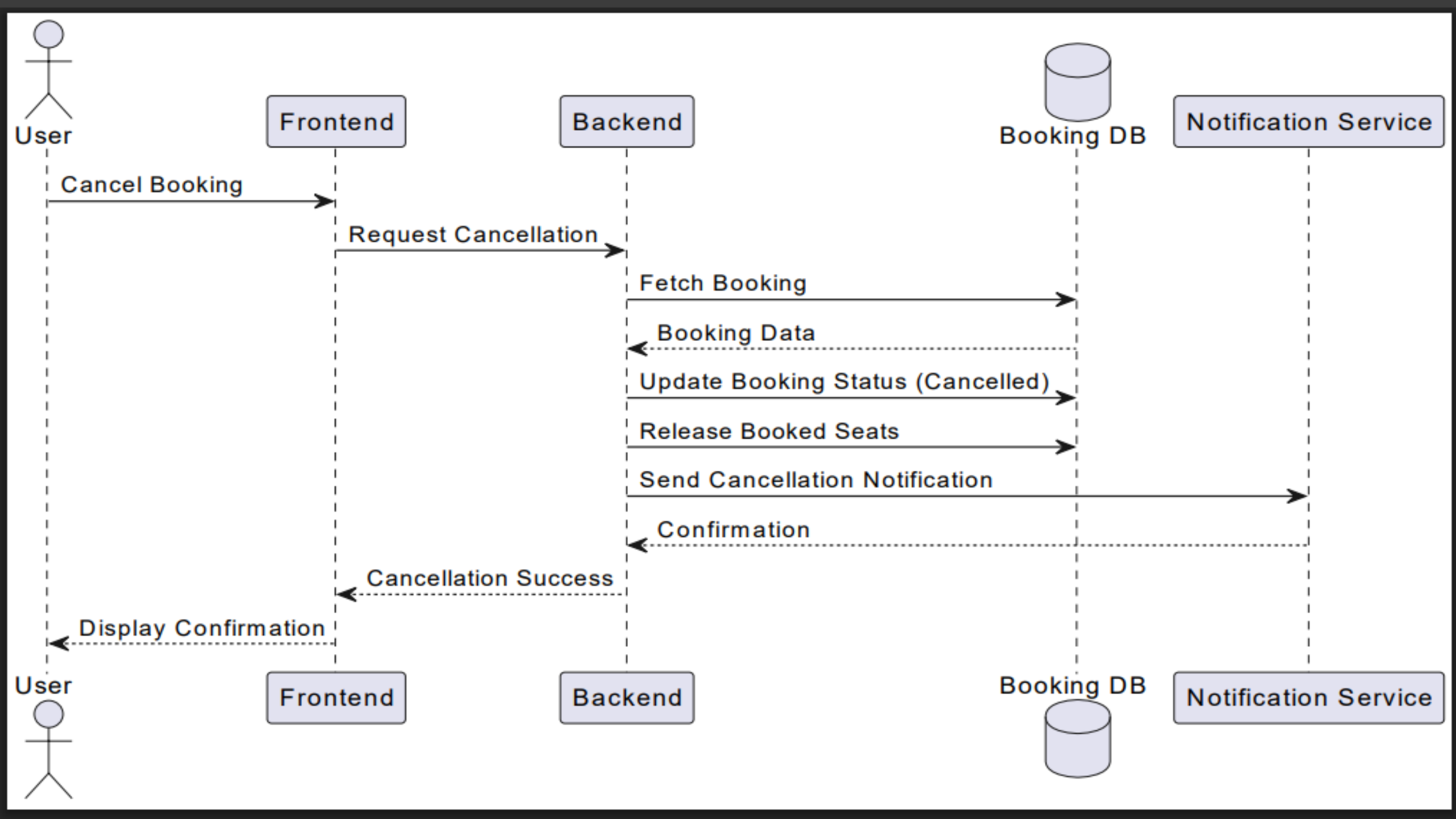
E1.3. The system notifies the user that personalised recommendations are temporarily unavailable.

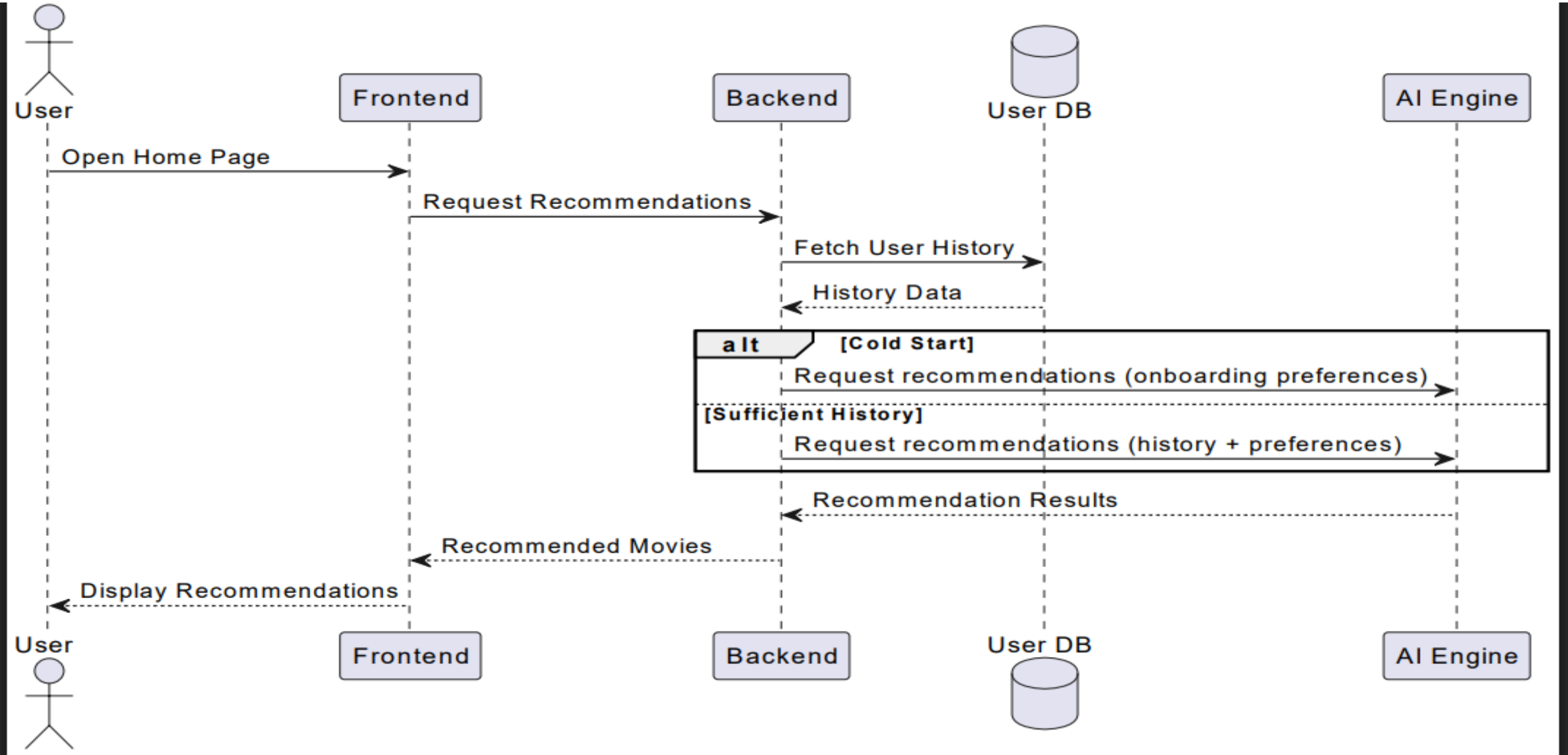
E1.4. The system displays a fallback list of currently featured or top-rated movies sourced from the Cinélux database.


Cinélux System
Register
Login
Browse Movies
View Movie Details
Make Booking
Cancel Booking
View Booking History
Get Movie Recommendations
Add Movie
Edit Movie
Delete Movie
Manage Showtimes
Manage Halls
View Reports
Validate Ticket
User
Admin
Staff
«include»
Send Notification (Internal Step)
«include»
Generate QR (Internal Step)
«include»
Prepare AI Request (Internal)
«include»
Process AI Response (Internal)
Notification Service
AI Engine


Generate QR Ticket Flow
User
Frontend
Backend
Booking DB
User Requests QR
View Ticket (bookingID)
Request Ticket Data
Fetch Booking
Get Booking Details
Booking Info
Generate QR
Build QR Payload
(bookingID, seat, showtime)
Generate QR Code
Return QR
QR Code
Display QR Ticket
User
Frontend
Backend
Booking DB


User
Frontend
Backend
Booking DB
Payment Service
Notification Service
Start Booking
Request Movie Details
Fetch Showtimes + Seats
Data
Display Seats
loop
[For each seat selected]
Select Seat
Hold Seat Temporarily
Mark Seat as Temporarily Held
Confirm Booking
Finalize Booking
Simulate Payment
Payment Result
Create Booking Record
Mark Seats as Booked
Send Booking Notification
Confirmation
Booking Success + QR
Display QR Ticket
User
Frontend
Backend
Booking DB
Payment Service
Notification Service


User
Frontend
Backend
Booking DB
Notification Service
Cancel Booking
Request Cancellation
Fetch Booking
Booking Data
Update Booking Status (Cancelled)
Release Booked Seats
Send Cancellation Notification
Confirmation
Cancellation Success
Display Confirmation
User
Frontend
Backend
Booking DB
Notification Service


User
Frontend
Backend
User DB
AI Engine
Open Home Page
Request Recommendations
Fetch User History
History Data
alt
[Cold Start]
Request recommendations (onboarding preferences)
[Sufficient History]
Request recommendations (history + preferences)
Recommendation Results
Recommended Movies
Display Recommendations
User
Frontend
Backend
User DB
AI Engine